

Why Log Normalisation Is Important in Cybersecurity

Raw log files hold a lot of valuable information, but they are often difficult to analyse because different systems record events in different formats. Linux systems typically use syslog, Windows relies on Event Logs, and many applications output JSON or CSV files instead. Some logs capture details such as the username, IP address, and action performed, while others omit several of these fields entirely. Inconsistent character encodings, such as UTF-8 versus Latin-1, add further parsing challenges. There is also no shared standard for severity levels or event types, which makes it hard to compare events coming from multiple sources.

Log normalisation addresses these problems by converting varied log formats into one consistent structure. Once every log follows the same schema, filtering and searching for specific events becomes far simpler — an analyst can, for instance, query `event_type == 'AUTH'` to pull authentication events from many different systems at once. Normalisation also makes it easier to correlate activity across sources, such as linking a firewall block to a failed login attempt. It strengthens visualisation through dashboards and graphs, and it prepares data for machine learning by keeping features consistent and well organised.

Over-normalising, however, introduces its own risks. Discarding the original raw log message can erase forensic details that investigators later need. Collapsing every kind of error into a single generic `ERROR` category can likewise hide meaningful information about what actually happened, and an overly rigid schema may strip out custom fields that matter for particular applications.

Consider analysing login attempts across ten different systems, each recording timestamps in its own format — `MMM DD HH:MM:SS`, `YYYY-MM-DDTHH:MM:SSZ`, or Unix epoch time. Without normalisation, each format has to be parsed separately. Once every timestamp is converted to UTC, a single query such as `action='LOGIN' AND timestamp > '2026-07-20'` can run across all sources at once.

KEY TAKEAWAY

Normalise logs enough to make analysis easier, but always keep the original raw log so important details are not lost.